Bias Inheritance in Neural-Symbolic Closure Discovery Under Function-Class Mismatch

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Abstract

We study closure discovery in nonlinear reaction-diffusion systems when the governing PDE structure is known but the constitutive laws remain unknown. Our goal is to recover the diffusion and reaction closures from spatiotemporal observations, while avoiding the usual failure mode in which low residuals or short-horizon predictions are mistaken for correct physical recovery. We propose a neural-symbolic pipeline with three stages: weak-form-driven hybrid numerical constitutive recovery, restricted symbolic compression, and forward re-simulation validation. Stage 1 learns numerical surrogates for the unknown closures under positivity and smoothness constraints using a weak-form-driven hybrid objective; Stage 2 compresses the learned surrogates into restricted polynomial, rational, and saturation-style families; Stage 3 reinserts the symbolic closures into the PDE and evaluates unseen rollout behavior. Experiments on 1D reaction-diffusion systems show two distinct regimes. In matched-library settings, weak polynomial baselines behave like correctly specified reference estimators and should not be expected to be uniformly outperformed by neural surrogates. In function-class-mismatch settings, neural surrogates provide flexible numerical closures that can be compressed into compact symbolic laws with small additional surrogate error and little additional rollout degradation on the reported benchmarks. However, symbolic compression does not automatically repair constitutive bias: across the reported clean, noisy, and sparse observation settings, the symbolic true closure error closely tracks the neural true closure error, and the bias inheritance ratio remains near one. On the studied benchmarks, these results indicate that the dominant bottleneck appears to lie in Stage 1 numerical constitutive recovery rather than in sym-

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bolic compression, and that closure claims should be supported by explicit forward validation rather than residual minimization alone. Code, manuscript source, and selected paper-facing artifacts are available at https://github.com/ToughClimb/closure-bias-inheritance.

Keywords: closure discovery, weak form, symbolic regression, reaction-diffusion, identifiability

1. Introduction

Many scientific systems are governed by partial differential equations whose large-scale structure is known while crucial constitutive closures remain unavailable. This setting appears in transport, pattern formation, continuum modeling, data-driven computational mechanics, and data-driven closure modeling for fluid and multiscale systems [1–4]. In such problems, the conservation law or balance law is understood, but the diffusion, mobility, reaction, or source terms must still be inferred from data. The relevant scientific object is therefore not only a predictive simulator but the closure law itself.

Data-driven discovery of governing equations and sparse identification of PDEs have become central threads in scientific machine learning [5–10]. Related hidden-physics inverse solvers, physics-informed inverse solvers, sparse optimization frameworks with physical constraints, and neural model-discovery pipelines further expanded this line of work [11–17]. Symbolic regression is especially attractive in this setting because the desired output is often an interpretable constitutive law rather than only a predictive black-box surrogate [18–22].

This paper studies closure discovery in nonlinear reaction-diffusion systems of the form

$$u_t = \partial_x \left(D(u)u_x\right) + R(u), \tag{1}$$

where the PDE structure is known but the constitutive relations $D(u)$ and $R(u)$ are unknown. Reaction-diffusion systems are canonical models for pattern formation and morphogenesis [23–25]. Given spatiotemporal observations of $u(x,t)$, we seek to recover these closures in a form that is numerically valid, scientifically interpretable, and verifiable by forward simulation.

The main challenge is that closure discovery is an inverse problem rather than a pure prediction problem. A small residual on observed trajectories, or even a small short-horizon rollout error, does not by itself imply correct constitutive recovery. Under limited excitation, different closure pairs may compensate each other on the observed support and produce similar trajectory-level behavior. This creates a gap between fitting the observed dynamics and identifying the true law. That concern is consistent with the broader inverse-problem literature, where limited informativeness and ill-posedness are central obstacles [26–28]. Closing that gap is the main methodological issue addressed in this work.

This perspective immediately changes how methods should be evaluated. In matched-library settings, where the true closure lies in a low-order polynomial dictionary, weak polynomial baselines can behave like correctly specified reference estimators and should be treated as strong comparators rather than weak strawmen [7, 8]. In mismatch settings, where the true closure lies outside that restricted library, flexible neural surrogates become relevant as numerical constitutive approximators. However, even then, a neural surrogate is not yet a scientific law. It still must be compressed into a symbolic form and then reinserted into the PDE to check whether the compressed law preserves the learned dynamics, in the spirit of recent neural-symbolic discovery workflows [15, 18–20].

Our paper therefore adopts a three-stage neural-symbolic view of closure discovery. Stage 1 learns numerical constitutive surrogates with a weak-form-driven hybrid objective. Stage 2 compresses those surrogates into restricted symbolic families. Stage 3 validates the symbolic laws by forward re-simulation on unseen initial conditions. This decomposition lets us ask a sharper question than "does the method get low loss?": what part of the pipeline is actually responsible for constitutive error?

The answer supported by our theory and experiments on these benchmarks is the following. Symbolic compression can preserve learned constitutive behavior with very small additional approximation error and almost no rollout degradation, but it does not automatically repair constitutive bias inherited from the neural surrogate. Across the reported clean, noisy, and sparse observation settings, the symbolic true closure error closely tracks the neural true closure error, and the bias inheritance ratio remains near one. On the studied benchmarks, the main bottleneck in neural-symbolic closure discovery therefore appears in Stage 1 numerical constitutive recovery rather than Stage 2 symbolic compression.

The contributions of this paper are fourfold.

1. We formulate closure discovery under known PDE structure as a weak-form-driven inverse problem with explicit attention to identifiability, excitation coverage, and function-class mismatch.

2. We propose a neural-symbolic pipeline that separates numerical constitutive recovery, restricted symbolic compression, and forward validation by PDE re-simulation.

3. We provide a theory spine explaining why low residual does not guarantee true closure recovery, why matched-library baselines can be unusually strong in the correctly specified regime, and why symbolic compression preserves rather than repairs constitutive bias.

4. We benchmark the method against strong and weak polynomial baselines across clean, noisy, and sparse regimes, supporting the view that the central mechanism in the reported settings is bias preservation through compression rather than symbolic correction.

2. Problem Formulation

We consider one-dimensional reaction-diffusion systems on the periodic domain $\Omega = [0, 1)$:

$$u_t = \partial_x (D(u)u_x) + R(u), \qquad x \in \Omega, \quad t \in [0, T]. \tag{2}$$

The field $u(x, t)$ is observed, while the closure pair

$$f = (D, R) \tag{3}$$

is unknown. The task is to recover $D(u)$ and $R(u)$ from spatiotemporal data generated by multiple trajectories with varying initial conditions.

We study this problem under two regimes.

- **Matched-library regime.** The true closure lies in a restricted polynomial family. This is the setting of Case A and Case B and serves as a correctly specified reference regime.
- **Function-class-mismatch regime.** The true closure lies outside the restricted polynomial library. This is the setting of Case Exp and is the main stress test for the neural-symbolic pipeline.

The data consist of finite trajectory sets

$$\mathcal{D} = \left\{u^{(m)}(x, t_n)\right\}_{m=1}^{M}, \tag{4}$$

possibly corrupted by Gaussian noise and/or spatial-temporal downsampling. The objective is not only to fit these trajectories, but to recover closures that remain valid under symbolic compression and forward rollout on unseen initial conditions.

We therefore evaluate every method at three levels.

1. **Trajectory level:** weak residual and unseen rollout error.
2. **Closure level:** relative error of $D(u)$ and $R(u)$ on a common support grid.
3. **Neural-symbolic level:** error between the symbolic closure and the learned neural surrogate, plus the post-compression rollout error.

For the closure-level metrics, we evaluate both true and recovered closures on a common support grid $\{u_\ell\}_{\ell=1}^{L} \subset \mathcal{U}$ over the relevant state interval and report

$$\mathrm{ErrD} = \frac{\left(\sum_{\ell=1}^{L} |D(u_\ell) - \widehat{D}(u_\ell)|^2\right)^{1/2}}{\left(\sum_{\ell=1}^{L} |D(u_\ell)|^2\right)^{1/2} + 10^{-12}}, \tag{5}$$

$$\mathrm{ErrR} = \frac{\left(\sum_{\ell=1}^{L} |R(u_\ell) - \widehat{R}(u_\ell)|^2\right)^{1/2}}{\left(\sum_{\ell=1}^{L} |R(u_\ell)|^2\right)^{1/2} + 10^{-12}}. \tag{6}$$

For unseen rollout, we compare full predicted and true trajectory arrays on shared initial conditions and report the relative trajectory error

$$\varepsilon_{\mathrm{roll}} = \frac{\|u_{\mathrm{true}} - u_{\mathrm{pred}}\|_2}{\|u_{\mathrm{true}}\|_2 + 10^{-12}}. \tag{7}$$

The central question of the paper is not whether a model can reduce a training loss, but whether the recovered closure survives the full chain

$$\begin{aligned}\text{observations} &\rightarrow \text{numerical surrogate} \rightarrow \text{symbolic compression} \\ &\rightarrow \text{forward validation.}\end{aligned} \tag{8}$$

3. Method

3.1. Weak-form constitutive recovery

Direct strong-form differentiation is notoriously brittle under noise and sparse observations. We therefore train the numerical closure surrogate in a *space-weak* form. For a spatial test function $\phi = \phi(x)$ and periodic boundary conditions, (1) yields, for each observation time t_n,

$$\int_\Omega \phi u_t \, dx + \int_\Omega D(u) u_x \phi_x \, dx - \int_\Omega \phi R(u) \, dx = 0. \tag{9}$$

Given a set of test functions $\{\phi_k\}$ and discrete trajectories, we define the weak residual by projecting the observed field onto (9) at each saved time step. In the current implementation, the time derivative u_t is approximated by a central difference on the saved temporal grid, while the spatial operator is handled in weak form. This avoids differentiating the data twice in space and provides a central physics-consistent component of the hybrid training signal. Related weak or integral formulations have been used to improve robustness of equation discovery under noisy or incomplete data [10, 29, 30].

3.2. Numerical closure surrogate

We use a compact neural surrogate with two scalar branches, one for diffusion and one for reaction:

$$\widehat{D}(u), \qquad \widehat{R}(u). \tag{10}$$

The diffusion branch is constrained to remain positive by construction, while both branches are regularized for smoothness on the observed state range. In the current implementation the main backbone is a small residual MLP built on explicit polynomial features.

The training objective combines weak-form consistency with auxiliary rollout and regularization terms:

$$\mathcal{L} = \mathcal{L}_{\text{weak}} + \alpha\mathcal{L}_{\text{roll}} + \beta\mathcal{L}_{\text{mass}} + \gamma\mathcal{L}_{\text{strong}} + \eta\mathcal{L}_{\text{anchor}} + \lambda\mathcal{L}_{\text{reg}}. \tag{11}$$

Here $\mathcal{L}_{\text{mass}}$ explicitly reweights the spatially averaged balance relation already implied by the constant test function in the weak form, $\mathcal{L}_{\text{strong}}$ provides an auxiliary pointwise consistency term only in the synthetic regime, $\mathcal{L}_{\text{anchor}}$ softly anchors the reaction branch at the lower state boundary, and $\mathcal{L}_{\text{reg}}$ penalizes excessive variation of the learned closures. The conceptual pipeline remains weak-form driven: the auxiliary rollout, reweighted balance, anchor, and synthetic strong-form terms regularize the numerical surrogate, but the central identifiability question is still whether Stage 1 recovers the constitutive laws rather than merely fitting trajectories.

3.3. Restricted symbolic compression

Once the numerical closures are learned, we evaluate them on a dense support grid in u and fit restricted symbolic families to the sampled constitutive curves. This stage is intentionally framed as *symbolic compression of learned constitutive surrogates*, not as unrestricted symbolic discovery directly from raw trajectories.

The candidate families are deliberately small and interpretable. For diffusion we consider low-order polynomials, a low-order rational family, exponential-decay forms, and saturation forms. For reaction we consider low-order polynomials, a low-order rational family, and two one-factor nonlinear families of the forms $u \exp(\cdot)$ and $u/(1+\cdot)$. For each closure, we select the lowest-complexity candidate whose fit error lies within a small tolerance of the best candidate. This gives a symbolic pair

$$f_{\mathrm{sym}} = \left(D_{\mathrm{sym}}, R_{\mathrm{sym}}\right) \tag{12}$$

that is compact enough to inspect while remaining faithful to the neural surrogate.

This restricted library is a feature, not a bug. The goal of Stage 2 is not exhaustive symbolic search over arbitrary formulas. The goal is to test whether a learned numerical constitutive surrogate can be compressed into a compact family without materially changing its induced dynamics.

3.4. Forward validation

The symbolic stage only counts if the compressed closure can be reinserted into the PDE and rolled forward successfully. We therefore validate both the learned numerical surrogate and the compressed symbolic closure by forward simulation on shared unseen initial conditions. This produces two rollout metrics:

$$\varepsilon_{\mathrm{roll}}^{\mathrm{neural}}, \qquad \varepsilon_{\mathrm{roll}}^{\mathrm{sym}}. \tag{13}$$

Their comparison is essential: if the symbolic closure substantially degrades rollout, then the compression stage failed even if curve fitting looked accurate in state space.

4. Theory

The theory in this paper is intended as a mechanism-level explanation of the observed behavior, not as a full uniqueness-and-convergence theory for nonlinear PDE inverse problems. Its role is to justify the weak-form objective, explain when polynomial baselines are especially strong in the correctly specified regime, and formalize why symbolic compression preserves but does not repair constitutive bias.

Unless stated otherwise, the closure-space norms below are taken in a common norm on the scalar constitutive functions. When a norm is restricted to the bounded state interval visited by the trajectories, we write it explicitly as an $L^\infty(\mathcal{U})$ norm. In the empirical evaluation, the reported closure metrics are relative discrete $L^2(\mathcal{U})$ errors on a common support grid, which is consistent with the topology induced by the weak-form objective and the rollout comparisons used in the paper.

4.1. Weak-form consistency

Proposition 1 (Space-weak consistency)**.** *Assume*

$$u \in L^2(0,T;H^1(\Omega)), \qquad u_t \in L^2(0,T;H^{-1}(\Omega)). \tag{14}$$

Then for any spatial test function $\phi \in H^1(\Omega)$,

$$\int_0^T \langle u_t, \phi \rangle \, dt + \int_0^T \int_\Omega D(u) u_x \phi_x \, dx dt - \int_0^T \int_\Omega R(u)\phi \, dx dt = 0. \tag{15}$$

Under periodic or no-flux boundaries,

$$\frac{d}{dt} \int_\Omega u \, dx = \int_\Omega R(u) \, dx. \tag{16}$$

This proposition legitimizes the weak residual and the reweighted mass-balance penalty. The second identity is obtained by choosing $\phi \equiv 1$, so the diffusion contribution vanishes and only the reaction-driven mass balance remains. In the implementation, $\mathcal{L}_{\text{mass}}$ isolates this scalar relation and gives it additional weight on top of the general weak residual. The result should be read as a consistency statement for the space-weak training objective. It is not intended as a complete space-time weak-solution definition.

4.2. Matched-library identification

Proposition 2 (Discrete projected identification under sufficient excitation)**.** *Suppose the true closures lie in finite dictionaries:*

$$D(u) = \sum_{i=1}^{p_D} a_i \psi_i(u), \qquad R(u) = \sum_{j=1}^{p_R} b_j \chi_j(u). \tag{17}$$

After weak-form projection and discrete quadrature, the identification problem becomes

$$\Phi\theta = Y, \tag{18}$$

where θ stacks the coefficients. If the Gram matrix

$$G = \Phi^\top \Phi \tag{19}$$

is positive definite, then the least-squares estimator is unique.

Proof sketch. The discrete projected fit is the least-squares problem

$$\min_{\theta} \|\Phi\theta - Y\|_2^2. \tag{20}$$

Its normal equations are $G\theta = \Phi^\top Y$ with $G = \Phi^\top \Phi$. If G is positive definite, then it is invertible, so the least-squares estimator is unique and equals $\theta = G^{-1}\Phi^\top Y$.

This explains why weak polynomial baselines are so strong in Case A and Case B. The important point is not merely that the model is linear in coefficients, but that excitation coverage determines whether the induced design matrix has informative, nondegenerate columns. This is a uniqueness statement for the projected coefficient fit, not a full statistical consistency theorem for PDE closure recovery.

4.3. Low residual does not imply true closure recovery

Proposition 3 (Non-identifiability under limited excitation)**.** *Let $\mathcal{A}_{\mathcal{T}}(D, R)$ denote the weak-form observation operator induced by a finite trajectory set $\mathcal{T}$. If $\mathcal{A}_{\mathcal{T}}$ is not injective on the admissible closure class, then there exist nonzero perturbations $(\delta D, \delta R)$ such that*

$$\mathcal{A}_{\mathcal{T}}(D + \delta D, R + \delta R) \approx \mathcal{A}_{\mathcal{T}}(D, R). \tag{21}$$

Hence low weak residual on the observed trajectories does not imply unique or correct recovery of the true closure pair.

Here "$\approx$" means approximate equality in the induced observation norm or, equivalently in practice, a comparably small weak residual on the observed trajectory set. This proposition formalizes the central inverse-problem warning behind the paper. It explains why low weak loss and small short-horizon rollout error can coexist with large closure error, especially when the observed data occupy only a narrow state interval or have weak gradient and curvature content.

4.4. Approximation bias under function-class mismatch

Proposition 4 (Approximation bias)**.** *Let $\mathcal{F}$ be the hypothesis class used by the identifier. If the true closure f^* does not belong to $\mathcal{F}$, define the best-in-class target*

$$f^\dagger = \arg\min_{f \in \mathcal{F}} \mathcal{L}(f). \tag{22}$$

Then the recovery error satisfies

$$\|f^* - \widehat{f}\| \le \|f^* - f^\dagger\| + \|f^\dagger - \widehat{f}\|. \tag{23}$$

The first term is model-class bias and the second is identification or optimization error. This decomposition explains why the mismatch regime is the right place to justify a neural surrogate: its main value is reduced approximation bias relative to a misspecified restricted library.

4.5. Symbolic compression as bias-preserving distillation

Proposition 5 (Bias preservation under symbolic compression)**.** *Let f^* be the true closure, $\widehat{f}$ the learned numerical surrogate, and f_{sym} the compressed symbolic closure. Then*

$$\left| \|f^* - f_{sym}\| - \|f^* - \widehat{f}\| \right| \le \|\widehat{f} - f_{sym}\|. \tag{24}$$

Proof sketch. Apply the reverse triangle inequality to $a = f^* - f_{\text{sym}}$ and $b = f^* - \widehat{f}$:

$$\left| \|a\| - \|b\| \right| \le \|a - b\| = \|\widehat{f} - f_{\text{sym}}\|, \tag{25}$$

which is exactly (24).

Equation (24) is the core theoretical explanation of the benchmark results. If compression error is small, then the symbolic true error must remain close to the neural true error. In other words, the symbolic stage cannot create a large correction unless it first departs substantially from the surrogate it is meant to compress. The estimate is a direct application of the reverse triangle inequality to the pair of vectors $f^* - f_{\text{sym}}$ and $f^* - \widehat{f}$.

4.6. Rollout preservation under small closure perturbations

Remark 1 (Finite-time rollout stability)**.** *Let u_1 and u_2 solve the same initial-value problem on $[0, T]$ with closure pairs $f_1 = (D_1, R_1)$ and $f_2 = (D_2, R_2)$. Under uniform parabolicity and bounded-Lipschitz assumptions on both closures, there exists a constant C_T such that*

$$\sup_{t\in[0,T]} \|u_1(t) - u_2(t)\|_{L^2} \le C_T \left(\|D_1 - D_2\|_{L^\infty(\mathcal{U})} + \|R_1 - R_2\|_{L^\infty(\mathcal{U})} \right), \tag{26}$$

where $\mathcal{U}$ is the bounded state range visited by the trajectories.

This estimate should be read as a standard finite-time stability template rather than a fully derived theorem under minimal assumptions. It connects small closure-space compression error to small additional rollout degradation over finite horizons and explains why symbolic rollout almost matches neural rollout in the experiments.

4.7. Unified error decomposition

Combining the approximation-bias and compression statements gives the practical decomposition

$$
\begin{aligned}
\|f^* - f_{\mathrm{sym}}\| \le \; & \underbrace{\|f^* - f^\dagger\|}_{\text{model-class bias}} \\
& + \underbrace{\|f^\dagger - \widehat{f}\|}_{\text{identification and optimization error}} \\
& + \underbrace{\|\widehat{f} - f_{\mathrm{sym}}\|}_{\text{symbolic compression error}} .
\end{aligned}
\tag{27}
$$

The experiments in this paper indicate that, on the reported benchmarks, the third term is comparatively small while the first two dominate. On these benchmarks, this is why the symbolic stage is not the main bottleneck.

5. Experiments

5.1. Experimental protocol

All main experiments are performed on 1D periodic reaction-diffusion systems. We evaluate three representative cases:

- **Case A:** a matched-library reference case with

$$D(u) = 0.01 + 0.05u, \qquad R(u) = u(1 - u).$$

- **Case B:** a second matched-library case with

$$D(u) = 0.01 + 0.03u^2, \qquad R(u) = u - 1.5u^2 + 0.5u^3.$$

- **Case Exp:** a function-class-mismatch stress test with

$$D(u) = 0.01 + 0.035(1 - e^{-2.5u}), \qquad R(u) = u(1.1e^{-1.4u} - 0.22).$$

We compare four method families:

1. strong polynomial baseline,
2. weak polynomial baseline,

Case	Setting	Method	ErrD	ErrR	Unseen
Case A	clean	strong_poly	4.098e−04 ± 1.144e−04	2.113e−03 ± 8.855e−04	1.804e−05 ± 5.128e−06
Case A	clean	weak_poly	8.151e−03 ± 1.047e−03	3.090e−02 ± 9.034e−03	8.307e−04 ± 1.348e−04
Case A	clean	neural	6.926e−02 ± 4.007e−03	3.128e−01 ± 1.892e−02	4.828e−03 ± 3.137e−04
Case A	clean	neural+symbolic	6.853e−02 ± 4.266e−03	3.127e−01 ± 1.894e−02	4.701e−03 ± 3.516e−04
Case B	clean	strong_poly	4.018e−04 ± 1.088e−04	2.597e−03 ± 1.092e−03	1.651e−05 ± 4.630e−06
Case B	clean	weak_poly	8.030e−03 ± 6.990e−04	3.701e−02 ± 8.150e−03	9.556e−04 ± 1.252e−04
Case B	clean	neural	1.003e−01 ± 5.687e−03	2.047e−01 ± 1.727e−02	3.949e−03 ± 3.374e−04
Case B	clean	neural+symbolic	1.001e−01 ± 5.650e−03	2.036e−01 ± 1.756e−02	3.727e−03 ± 3.585e−04
Case Exp	clean	strong_poly	4.609e−02 ± 2.701e−04	4.270e−01 ± 1.115e−01	3.146e−03 ± 2.813e−04
Case Exp	clean	weak_poly	5.230e−02 ± 1.786e−03	8.712e−02 ± 1.598e−02	2.738e−03 ± 8.345e−05
Case Exp	clean	neural	4.772e−02 ± 7.372e−04	4.400e−01 ± 8.150e−03	3.212e−03 ± 7.913e−05
Case Exp	clean	neural+symbolic	4.765e−02 ± 8.388e−04	4.400e−01 ± 8.151e−03	3.226e−03 ± 9.908e−05

Table 1: Representative baseline comparison. In matched-library settings, polynomial baselines are exceptionally strong. In mismatch settings, the neural surrogate serves as a flexible numerical constitutive approximator, while the symbolic stage largely preserves its behavior.

3. neural numerical surrogate,

4. neural surrogate followed by symbolic compression.

Unless otherwise stated, the neural benchmarks use 8 training trajectories, 4-mode random Fourier initial conditions, and the simulation grid $n_x = 64$ with integrator step size $\Delta t = 10^{-4}$ up to final time $T = 0.1$, with snapshots saved every 10 steps. Models are trained for 150 epochs. A "5% noise" setting means additive Gaussian perturbations with standard deviation $0.05\,\mathrm{std}(u)$. A "sparse $x + t$" setting means spatial stride 2 and temporal stride 2. Each reported value is averaged across three random seeds.

5.2. Baseline map: matched library versus mismatch

Table 1 summarizes representative settings. The main qualitative pattern is immediate. In the matched-library regime, strong and weak polynomial baselines are extremely competitive and behave like correctly specified reference estimators. In the mismatch regime, the neural surrogate becomes relevant as a flexible numerical approximation layer, but its advantage is metric-dependent and does not automatically translate into better rollout. In fact, for Case Exp under clean data, the weak polynomial baseline still gives the best rollout and the smallest reaction error, while the strong polynomial baseline remains competitive on diffusion. Function-class mismatch therefore does not by itself guarantee a uniform neural advantage under the current Stage 1 objective.

Case	Regime	Bin coverage	Weak diffusion energy	Weak loss	ErrD	ErrR	Unseen
Case A	low excitation	1.250e−01 ± 0.000e+00	1.091e+00 ± 1.270e−01	1.879e−05 ± 3.018e−06	1.280e−01 ± 9.278e−03	1.289e+00 ± 3.130e−02	5.333e−04 ± 8.114e−05
Case A	high excitation	1.000e+00 ± 0.000e+00	1.158e+02 ± 1.368e+01	9.471e−04 ± 1.602e−04	5.488e−02 ± 1.125e−02	3.673e−01 ± 2.049e−02	3.879e−03 ± 3.066e−04
Case B	low excitation	1.250e−01 ± 0.000e+00	1.822e+00 ± 2.590e−01	1.616e−05 ± 2.869e−06	2.984e−01 ± 1.291e−02	1.031e+00 ± 8.982e−02	4.744e−04 ± 8.218e−05
Case B	high excitation	1.000e+00 ± 0.000e+00	2.046e+02 ± 2.852e+01	3.691e−04 ± 5.804e−05	8.566e−02 ± 4.676e−03	2.829e−01 ± 3.448e−02	3.310e−03 ± 1.639e−04
Case Exp	low excitation	1.250e−01 ± 0.000e+00	2.411e+00 ± 2.782e−01	1.259e−05 ± 1.876e−06	1.706e−01 ± 1.758e−02	8.154e−01 ± 1.803e−02	2.743e−04 ± 1.822e−05
Case Exp	high excitation	1.000e+00 ± 0.000e+00	2.376e+02 ± 2.818e+01	3.872e−04 ± 7.170e−05	4.657e−02 ± 1.591e−03	4.052e−01 ± 2.217e−02	2.671e−03 ± 4.586e−05

Table 2: Excitation benchmark with three seeds. Narrow low-excitation training sets achieve very small weak losses but much worse closure recovery. The observed trajectory fit alone is therefore not a reliable identifiability diagnostic.

5.3. Excitation coverage and identifiability

Table 2 makes the identifiability issue explicit. For computational efficiency, this excitation benchmark uses a lighter protocol with $n_x = 48$, $T = 0.05$, snapshots saved every 5 steps, and 6 training trajectories. We compare low-excitation and high-excitation training sets by shrinking or expanding the amplitude range of the random initial conditions. The low-excitation regime occupies only a narrow state interval and produces much weaker gradient and diffusion-energy statistics. In all cases, this regime attains a *smaller* weak loss and even a substantially smaller short-horizon unseen rollout error, yet the recovered closures are substantially worse. Because the signal energy is also much smaller in the low-excitation regime, these weak-loss values should be read as absolute residual magnitudes rather than normalized cross-dataset scores. The conclusion is the one predicted by Proposition 3: low residual is not enough when the observation operator is weakly excited.

5.4. Symbolic compression preserves surrogate behavior

The core symbolic benchmark is summarized in Table 3 and Figure 2. Across the reported settings, the symbolic surrogate error remains much smaller than the neural true error. This indicates that the symbolic stage is primarily compressing the learned constitutive curves rather than inventing a different law. At the same time, the "noise 5%" rows indicate that the current weak-form-driven hybrid Stage 1 objective is not robust to this noise level on the present benchmarks: the neural true closure errors deteriorate sharply under observation noise, while the symbolic stage continues to inherit that constitutive bias rather than repair it. A dedicated six-level noise sweep across all three 1D cases (Figure 1) shows that these 5% rows are not isolated outliers. In each of the three reported cases, the mean closure errors rise sharply already between 1% and 3% noise and then flatten toward the 5% regime, so the practically relevant phenomenon is a protocol-specific degradation knee rather than a single pathological endpoint.

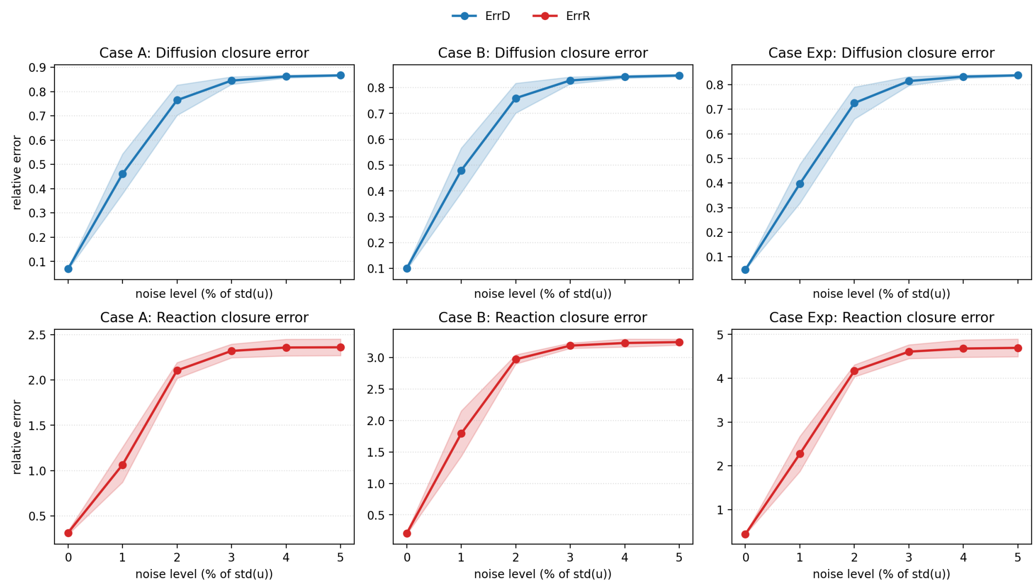

Figure 1: Stage 1 noise-sensitivity sweep across the three 1D benchmark cases. Mean±standard deviation over three seeds for the neural Stage 1 closure errors. In all cases, both diffusion and reaction recovery deteriorate sharply between 1% and 3% observation noise and then flatten toward saturated high-error regimes by 5%.

To quantify bias propagation, we define the bias inheritance ratios

$$\mathrm{BIR}_D = \frac{\mathrm{ErrD}^{\mathrm{sym,true}}}{\mathrm{ErrD}^{\mathrm{neural,true}}}, \qquad \mathrm{BIR}_R = \frac{\mathrm{ErrR}^{\mathrm{sym,true}}}{\mathrm{ErrR}^{\mathrm{neural,true}}}. \tag{28}$$

Values near one mean that symbolic compression preserves the constitutive bias already present in the neural surrogate. Values modestly below one indicate mild smoothing, not a systematic correction mechanism.

5.5. *PySR replacement check*

To test whether bias inheritance is merely an artifact of the restricted Stage 2 family, we replaced the symbolic-compression step with PySR [31] on the mismatch stress test (Case Exp) under the clean and "noise 5%" settings. The goal of this check is not to claim that PySR is the canonical Stage 2 engine, but to ask a sharper question: if Stage 2 is made substantially more expressive, does it actually repair the constitutive bias inherited from Stage 1? Table 4 indicates that, on this check, the answer remains no.

Case	Setting	Neural ErrD	Neural ErrR	Sym surrogate ErrD	Sym surrogate ErrR	BIR_D	BIR_R
Case A	clean	6.926e−02	3.128e−01	4.628e−03	6.453e−03	9.893e−01	9.999e−01
Case A	noise 5%	8.672e−01	2.361e+00	4.910e−04	4.212e−02	1.000e+00	1.002e+00
Case A	sparse x+t	5.046e−02	3.786e−01	1.518e−02	3.676e−03	9.239e−01	1.000e+00
Case B	clean	1.003e−01	2.047e−01	4.431e−03	2.370e−02	9.988e−01	9.942e−01
Case B	noise 5%	8.470e−01	3.248e+00	5.387e−04	7.419e−02	1.000e+00	1.007e+00
Case B	sparse x+t	8.184e−02	2.728e−01	5.720e−03	1.937e−02	9.969e−01	9.981e−01
Case Exp	clean	4.772e−02	4.400e−01	1.099e−03	2.363e−04	9.985e−01	1.000e+00
Case Exp	noise 5%	8.379e−01	4.695e+00	4.861e−04	1.877e−02	1.000e+00	9.992e−01
Case Exp	sparse x+t	7.167e−02	4.759e−01	2.282e−03	2.231e−04	9.987e−01	1.000e+00

Table 3: Representative symbolic compression summary. The symbolic stage stays close to the neural surrogate and the bias inheritance ratios remain near one, especially on the mismatch stress test.

Setting	Stage 2	Surrogate ErrD	Surrogate ErrR	Unseen	BIR_D	BIR_R
clean	restricted	1.099e−03 ± 4.705e−04	2.363e−04 ± 7.787e−05	3.226e−03 ± 9.908e−05	9.985e−01 ± 2.316e−03	1.000e+00 ± 1.649e−06
clean	PySR	5.638e−03 ± 2.965e−03	7.764e−03 ± 5.431e−03	3.302e−03 ± 1.041e−04	9.985e−01 ± 6.270e−03	9.958e−01 ± 4.358e−03
noise 5%	restricted	4.861e−04 ± 1.349e−04	1.877e−02 ± 9.893e−04	1.038e−01 ± 2.461e−03	1.000e+00 ± 3.239e−06	9.992e−01 ± 3.095e−04
noise 5%	PySR	5.019e−04 ± 1.329e−04	2.161e−02 ± 3.547e−03	1.038e−01 ± 2.499e−03	1.000e+00 ± 3.726e−06	1.000e+00 ± 5.670e−04

Table 4: Case Exp Stage 2 replacement check. On this mismatch stress test, replacing the restricted symbolic family with PySR leaves the qualitative conclusion unchanged: Stage 2 still preserves the constitutive bias inherited from the neural surrogate, especially under observation noise.

In the clean setting, PySR attains a small surrogate-fit error and a rollout of the same order as the restricted symbolic family, yet its true closure errors remain essentially inherited from the neural surrogate, with $\mathrm{BIR}_D \approx 9.985 \times 10^{-1}$ and $\mathrm{BIR}_R \approx 9.958 \times 10^{-1}$. Under 5% noise, the conclusion becomes even sharper on this mismatch stress test: both the restricted family and PySR remain at $\mathrm{BIR} \approx 1$, and their rollout errors stay near 1.04×10^{-1}. The main differences are complexity and, in the clean setting, somewhat looser surrogate compression rather than systematic correction. Averaged over three seeds, the PySR expressions have mean complexities of roughly $(17, 20.7)$ in the clean setting and $(17, 22)$ under 5% noise, versus $(5, 5)$ and $(3, 5)$ for the restricted family. This is consistent with the mechanism claim of the paper on the tested mismatch settings: bias inheritance is not caused solely by a weak symbolic library; it persists even when Stage 2 is replaced by a stronger symbolic search engine.

5.6. Cross-resolution anti-inverse-crime check

Table 5 evaluates the learned neural closure on a validation solver with a different grid and saved time step from the data-generation solver. The same-grid unseen rollout and the cross-grid rollout remain very close in all four observation settings. This does not prove full discretization independence, but it does indicate

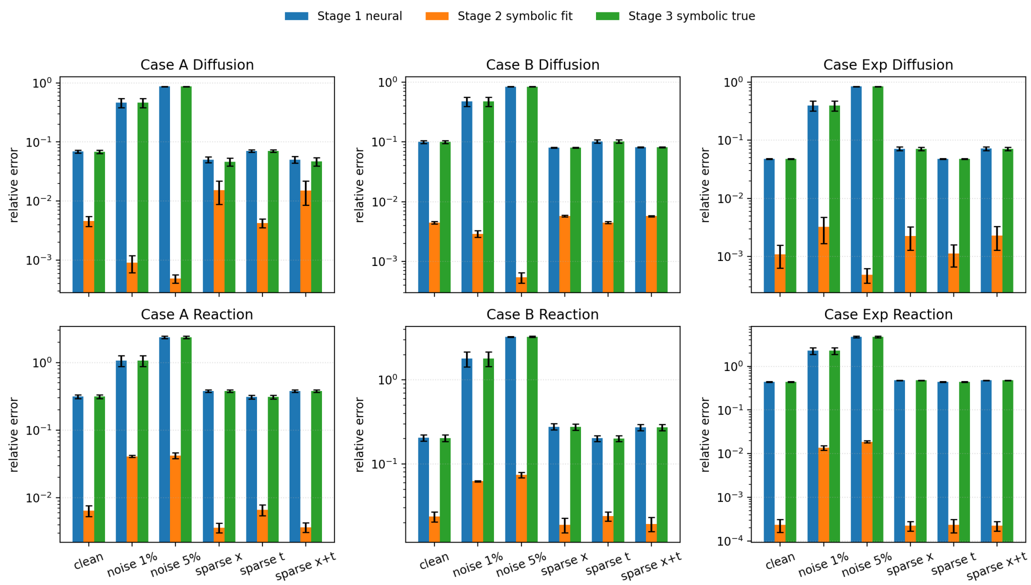

Figure 2: Neural-to-symbolic error propagation. Stage 2 compression error remains small, while Stage 3 true symbolic error tracks the Stage 1 neural true error. This is the empirical signature of bias preservation.

that the reported mismatch conclusions are not a trivial artifact of reusing exactly the same numerical configuration at training and validation time. A second pattern is also visible: in this benchmark, spatial downsampling is substantially more damaging than temporal downsampling, while temporal stride 2 alone leaves the errors almost unchanged relative to the $(1, 1)$ reference.

5.7. Rollout preservation and bias inheritance

Figure 3 shows two key dynamical consequences of symbolic compression on the reported settings. First, the symbolic rollout remains very close to the neural rollout across clean, noisy, and sparse settings. Second, the bias inheritance ratios for both diffusion and reaction remain near one. Together these results support the claim that symbolic compression is not correcting the constitutive law; it is distilling the numerical surrogate into a more compact representation while preserving its existing bias.

Observed stride (x, t)	ErrD	ErrR	Weak loss	Unseen	Cross-grid
$(1, 1)$	4.585e−02 ± 5.543e−04	4.081e−01 ± 1.701e−02	3.172e−04 ± 1.145e−05	2.606e−03 ± 2.861e−05	2.597e−03 ± 3.773e−05
$(2, 1)$	6.838e−02 ± 5.635e−03	4.377e−01 ± 2.040e−02	1.126e−03 ± 4.381e−04	3.678e−03 ± 2.677e−04	3.675e−03 ± 2.602e−04
$(1, 2)$	4.584e−02 ± 5.746e−04	4.082e−01 ± 1.685e−02	3.173e−04 ± 1.087e−05	2.606e−03 ± 2.908e−05	2.597e−03 ± 3.842e−05
$(2, 2)$	6.839e−02 ± 5.664e−03	4.378e−01 ± 2.029e−02	1.126e−03 ± 4.383e−04	3.678e−03 ± 2.689e−04	3.676e−03 ± 2.614e−04

Table 5: Cross-resolution anti-inverse-crime benchmark on Case Exp. This benchmark uses a separately generated dataset with a different grid and snapshot cadence from the default protocol. Training data are generated on a fine solver ($n_x = 64$, saved $\Delta t = 5\times10^{-4}$, saved horizon 5.0×10^{-2}), while validation uses a different grid and saved time step ($n_x = 48$, saved $\Delta t = 6 \times 10^{-4}$, last saved snapshot at 4.98×10^{-2}).

6. Discussion

The results support a more careful view of neural-symbolic closure discovery than the usual "neural beats symbolic" narrative. In matched-library settings, weak polynomial baselines are genuinely strong. This is not a pathological edge case for the neural model; it is consistent with what one should expect when the true closure lies inside the baseline hypothesis class and the induced weak-form system is sufficiently informative.

The mismatch regime is where the neural stage becomes scientifically meaningful. Even there, however, the role of the neural model is not to guarantee exact law discovery. Its role is to provide a flexible numerical constitutive surrogate when the restricted prior library is misspecified. The surrogate is therefore an intermediate object: useful, but not yet a validated scientific law.

The Stage 1 objective itself should also be interpreted carefully. A fixed-dataset ablation on Case Exp clean data (Appendix Table A.6) shows that a purely weak-form objective performs worst on reaction recovery, weak loss, and unseen rollout, while removing only the strong-form term yields the worst diffusion recovery. In other words, the implemented Stage 1 objective is not uniformly optimal across metrics, but neither is it reducible to weak-form training alone. This is why the paper describes Stage 1 as a weak-form-driven hybrid rather than as a pure weak-form learner.

This is why the symbolic stage must be interpreted carefully. The reported experiments show that symbolic compression is highly effective as a distillation step. It can reduce constitutive curves to compact symbolic forms with very small additional surrogate error and almost no extra rollout degradation on the tested settings. But once that compression error is small, it ceases to be the dominant source of error in those settings. The remaining constitutive bias is then dominated by mismatch and identification error from Stage 1. In this sense, the symbolic stage is not the main bottleneck in the current benchmarks; it is the place where

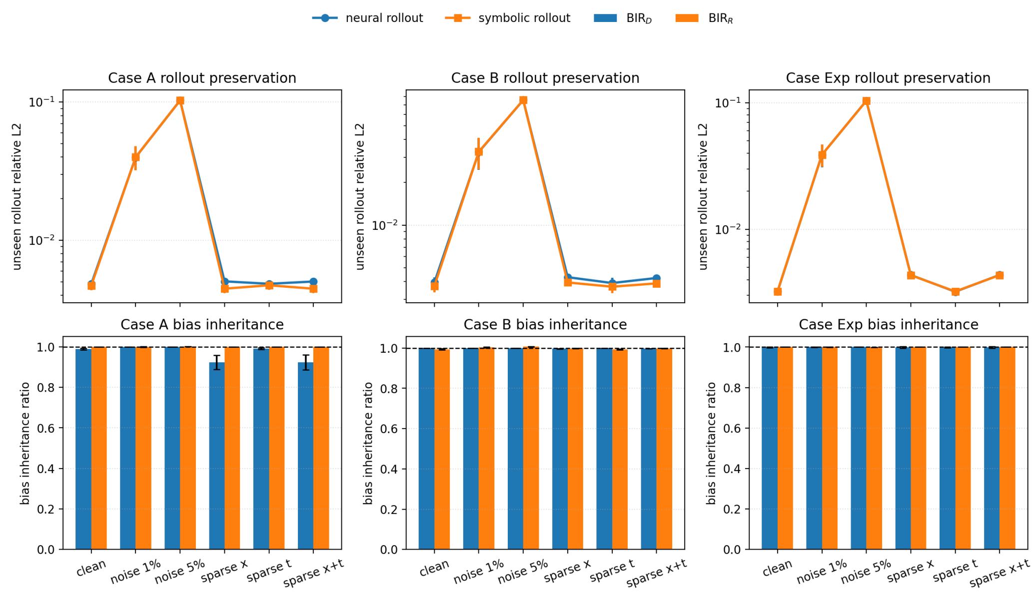

Figure 3: Rollout preservation and bias inheritance. Top: symbolic unseen rollout closely follows neural unseen rollout. Bottom: the bias inheritance ratios remain near one, showing that symbolic compression preserves rather than repairs Stage 1 constitutive bias.

the surrogate becomes interpretable, not the place where it becomes correct.

The noisy benchmarks sharpen this point rather than weaken it. Even with a weak-form-driven hybrid training signal, Stage 1 closure recovery deteriorates markedly under 5% observation noise, especially for the reaction term. This does not contradict the paper's thesis; it reinforces it. Weak-form training mitigates spatial differentiation brittleness, but it does not eliminate the inverse-problem difficulty of constitutive recovery under noisy, weakly informative observations. More broadly, optimization and conditioning difficulties are well known in physics-informed inverse solvers [32]. Table 4 makes the same point from the Stage 2 side: even when the restricted symbolic family is replaced by the much more expressive PySR engine, the noisy constitutive bias is still inherited rather than repaired on this mismatch stress test. Figure 1 sharpens the Stage 1 diagnosis across all three 1D cases: relative to the clean reference, the mean diffusion errors jump from $(6.93, 10.0, 4.77) \times 10^{-2}$ to $(4.61, 4.80, 3.97) \times 10^{-1}$ at 1% noise for Cases A, B, and Exp, while the reaction errors jump from $(3.13, 2.05, 4.40) \times 10^{-1}$ to $(1.06, 1.79, 2.27) \times 10^{0}$. Most of the remaining deterioration occurs by 2%–3% noise (nominal SNR about 34–30 dB), after which the curves largely saturate. This is better described as a protocol-specific degradation knee than as a universal critical threshold. Additional diagnostic ablations indicate that this knee should not be attributed to the space-weak residual alone. The current Stage 1 pipeline contains multiple explicit time-derivative channels, including the auxiliary strong residual and mass-balance terms in addition to the space-weak estimate of u_t, and the strongest noise sensitivity appears to come from that broader hybrid design rather than from the weak residual in isolation. The central scientific point is therefore structural: once Stage 1 has already lost the correct constitutive signal in the 1%–3% regime, a stronger Stage 2 engine such as PySR still inherits that degraded law in our current checks rather than repairing it. That shifts the scientific burden toward more robust Stage 1 numerical constitutive recovery under noise, potentially including fuller space-time weak formulations or other designs that reduce explicit time-derivative estimation, rather than increasingly elaborate symbolic post-processing.

These observations suggest a clear agenda for future work. The main difficulty is reliable numerical constitutive recovery under limited excitation, sparse observations, and misspecified prior structure. Better Stage 1 recovery may come from richer excitation design, stronger identifiability diagnostics, adaptive trajectory selection, improved regularization, or more problem-aware surrogate classes. In a few matched-library sparse settings, the diffusion error improves slightly relative to the clean reference, which is consistent with a mild implicit smoothing effect rather than a genuine information gain. In the current mismatch experiments, another asymmetry emerges: spatial coarsening is more harmful than temporal coarsening, suggesting that state-support and gradient information are more sensitive to spatial under-resolution than to a modest reduction in saved snapshot cadence. Appendix Figure A.4 makes that asymmetry explicit for Case Exp: temporal stride 2 alone stays very close to the clean reference, while spatial stride 2 accounts for almost all of the sparse-setting degradation. Once the numerical recovery problem is better controlled, the symbolic stage can remain simple and restricted.

The symbolic library is also intentionally narrow. It is designed to answer a specific question: can a learned surrogate be compressed into a compact family without changing its constitutive or dynamical behavior? It is not intended as a complete search space for nonlocal, singular, or history-dependent closures. If broader closure classes are needed, they should be introduced as controlled extensions of the same compress-and-reinsert protocol rather than as unvalidated formula mining.

The current study is intentionally narrow. Its main benchmark suite focuses on 1D periodic reaction-diffusion systems, synthetic data, and a controlled set of closure families. This keeps the mechanisms visible. Appendix Table A.7 adds a minimal 2D periodic dimensionality check, but it should be read as a preliminary stress test rather than a full higher-dimensional benchmark program. Extending the same logic to richer 2D/3D systems, transport-diffusion-reaction couplings, or more realistic observation models will introduce additional identifiability challenges, especially for anisotropic or multivariate closures, as is already familiar in data-driven closure modeling for fluid and continuum systems [3, 4]. That broader program is important future work, but it should be built on the present lesson: low residual is not enough, and symbolic compression should be judged by what it preserves.

7. Conclusion

We presented a weak-form-driven hybrid neural-symbolic framework for closure discovery in reaction-diffusion systems with known PDE structure and unknown constitutive laws. The framework separates numerical constitutive recovery, restricted symbolic compression, and forward validation by PDE re-simulation.

The main result is not that symbolic regression automatically repairs a biased neural surrogate. It does not on the reported benchmarks. Instead, the present experiments and theory support the claim that once symbolic compression error is small, the symbolic closure largely inherits the constitutive bias already present in the learned numerical surrogate. This is why symbolic rollout remains close to neural rollout and why the bias inheritance ratio stays near one across the reported clean, noisy, and sparse settings.

The practical conclusion is straightforward. In neural-symbolic closure discovery, the present benchmarks indicate that the main bottleneck appears in Stage 1 numerical constitutive recovery under limited excitation and function-class mismatch. A minimal 2D appendix benchmark is consistent with the same bias-inheritance mechanism under both clean and noisy mismatch settings, but it should not be read as a comprehensive higher-dimensional validation. Future progress should therefore focus on better identifiability, stronger validation protocols, and broader closure families, while keeping the symbolic stage restricted, interpretable, and explicitly validated by forward simulation.

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Appendix A. Implementation Details

Appendix A.1. Synthetic cases and numerical solver

All reported experiments use the 1D periodic solver in the accompanying codebase. Trajectories are generated by an RK4 time integrator applied to a conservative finite-volume-style discretization of

$$u_t = \partial_x \left(D(u)u_x\right) + R(u). \tag{A.1}$$

The three main benchmark cases are

$$\text{Case A:}\quad D(u) = 0.01 + 0.05u, \qquad R(u) = u(1-u), \tag{A.2}$$

$$\text{Case B:}\quad D(u) = 0.01 + 0.03u^2, \qquad R(u) = u - 1.5u^2 + 0.5u^3, \tag{A.3}$$

$$\begin{aligned}\text{Case Exp:}\quad & D(u) = 0.01 + 0.035(1 - e^{-2.5u}),\\ & R(u) = u(1.1e^{-1.4u} - 0.22).\end{aligned} \tag{A.4}$$

Case A and Case B are matched to the low-order polynomial baseline library. Case Exp is a smooth non-polynomial stress test for function-class mismatch.

Appendix A.2. Training and observation protocol

Unless otherwise stated, the baseline and symbolic benchmarks use 8 training trajectories, random smooth Fourier initial conditions with 4 modes, and the simulation grid

$$n_x = 64, \qquad \Delta t = 10^{-4}, \qquad T = 0.1, \tag{A.5}$$

with snapshots saved every 10 time steps. The excitation comparison in Table 2 uses a lighter protocol with $n_x = 48$, $T = 0.05$, snapshots saved every 5 steps, and 6 training trajectories. The cross-resolution benchmark in Table 5 uses the explicit fine and validation grids reported in its caption. The main surrogate is a two-branch MLP with hidden width 64, depth 2, and 150 epochs of Adam optimization at learning rate 10^{-3}.

Observation degradation is applied after simulation. A "5% noise" setting means additive Gaussian noise with standard deviation $0.05\,\mathrm{std}(u)$, followed by clipping to the admissible state range. A "sparse $x + t$" setting means spatial stride 2 and temporal stride 2 in the observed trajectories. Figure 1 further resolves the $0\%, 1\%, 2\%, 3\%, 4\%, 5\%$ noise response across all three 1D cases. All tables report mean±standard deviation over three random seeds unless stated otherwise.

Unless otherwise stated, both polynomial baselines use diffusion degree 2, reaction degree 3, and ridge regularization strength 10^{-8}. The weak polynomial baseline uses the same default test-function family as the paper-facing weak residual, namely 4 Fourier modes plus 4 bump functions. For the paper hybrid Stage 1 objective, the default weights are $\alpha = 0.1$, $\beta = 1$, $\gamma = 1$, $\eta = 1$, and $\lambda = 10^{-4}$. The main neural backbone uses tanh activations together with an explicit polynomial-residual parameterization: a degree-2 polynomial backbone for diffusion and a degree-3 polynomial backbone for reaction, each augmented by a small residual MLP. These settings are meant as a baseline reference rather than as a claim of exhaustive hyperparameter tuning. The public GitHub repository includes the training scripts, configuration dictionaries, and representative reproduction commands used to generate the paper tables; the main scientific point of the paper is the structural bias-inheritance mechanism, not extreme tuning of these specific weights.

Appendix A.3. Stage-1 objective ablation

Table A.6 reports a small fixed-dataset ablation on Case Exp under the clean protocol. The three compared objectives are: pure weak-form training, weak-form plus auxiliary terms but without the strong residual, and the full paper objective. The result is intentionally mixed. The full hybrid objective gives the best diffusion recovery, the smallest weak loss, and the best unseen rollout, while the weak-plus-auxiliary objective without the strong term gives the best reaction error. This is consistent with the main-text claim that the implemented Stage 1 learner is a weak-form-driven hybrid whose behavior is metric-dependent rather than uniformly dominant.

Objective	ErrD	ErrR	Weak loss	Unseen
weak only	9.072e−02 ± 2.879e−04	7.556e−01 ± 2.361e−03	1.151e−03 ± 1.077e−05	7.231e−03 ± 3.229e−04
weak + auxiliary, no strong	1.021e−01 ± 4.155e−04	2.480e−01 ± 4.824e−03	7.673e−04 ± 3.782e−06	5.637e−03 ± 6.639e−05
full paper hybrid	4.859e−02 ± 1.772e−04	4.369e−01 ± 7.874e−03	3.241e−04 ± 5.672e−06	3.319e−03 ± 1.465e−04

Table A.6: Small Stage 1 objective ablation on a fixed clean Case Exp dataset. Pure weak-form training is not competitive on this mismatch stress test. Removing the strong residual helps the reaction closure but still worsens diffusion recovery and rollout relative to the full paper objective.

Appendix A.4. Observation-degradation breakdown

Figure A.4 provides a complementary coarse view of the same Case Exp degradation mechanisms. The left panel compresses the dedicated noise sweep into representative clean, 1%, and 5% checkpoints while also showing the symbolic unseen rollout error. The right panel separates temporal from spatial downsampling: temporal stride 2 alone changes little, whereas spatial stride 2 drives nearly all of the sparse-setting penalty. This is consistent with the interpretation used in the main text, namely that the difficult part in this benchmark is preserving enough state-support and gradient information for Stage 1 recovery rather than merely preserving snapshot cadence.

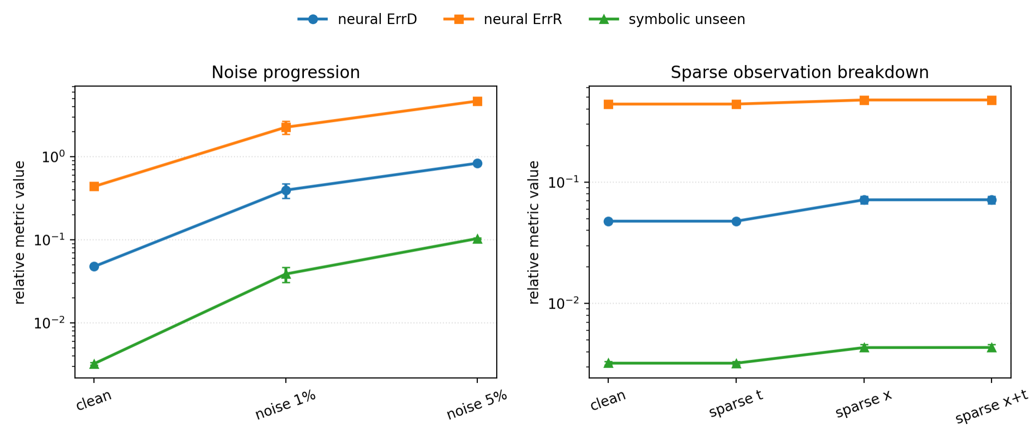

Figure A.4: Case Exp observation-degradation breakdown. Left: representative clean, 1%, and 5% noise checkpoints for the neural closure errors, together with the symbolic unseen rollout error. Right: temporal stride 2 alone leaves the mismatch benchmark almost unchanged, while spatial stride 2 accounts for nearly all of the sparse-setting degradation.

Appendix A.5. Minimal 2D dimensionality check

Table A.7 reports a deliberately small 2D periodic benchmark built from the same scalar closure families, but now with 32×32 grids, 4 training trajectories, and 120 optimization epochs. The goal is not to claim a fully mature 2D suite. It is to test whether the core mechanism survives a first dimensionality increase. The results are consistent with that same mechanism. On both the matched Case A and the mismatch Case Exp, the symbolic stage again stays very close to the learned neural surrogate, with BIR_D and BIR_R remaining near one. On the noisy 2D Case Exp stress test, Stage 1 deteriorates sharply, but Stage 2 still largely mirrors that degraded constitutive law rather than repairing it.

Case	Setting	Neural ErrD	Neural ErrR	Symbolic ErrD	Symbolic ErrR	Neural unseen	Symbolic unseen	BIR_D	BIR_R
Case A	clean	8.293e−02 ± 2.007e−03	3.188e−01 ± 2.663e−02	8.084e−02 ± 2.135e−03	3.188e−01 ± 2.664e−02	2.614e−03 ± 1.077e−04	2.007e−03 ± 1.064e−04	9.748e−01 ± 2.189e−03	9.999e−01 ± 5.449e−05
Case Exp	clean	8.669e−02 ± 1.458e−03	3.317e−01 ± 3.214e−02	8.573e−02 ± 1.524e−03	3.317e−01 ± 3.214e−02	2.092e−03 ± 8.974e−05	2.236e−03 ± 8.949e−05	9.889e−01 ± 9.944e−04	1.000e+00 ± 6.712e−06
Case Exp	noise 5%	5.719e−01 ± 2.168e−02	3.783e+00 ± 5.565e−02	5.719e−01 ± 2.178e−02	3.780e+00 ± 5.545e−02	2.437e−02 ± 1.270e−03	2.440e−02 ± 1.304e−03	1.000e+00 ± 3.079e−04	9.994e−01 ± 4.523e−05

Table A.7: Minimal 2D periodic dimensionality check with three seeds. The 2D benchmark is intentionally small and not tuned as aggressively as the 1D main suite, but it remains consistent with the same mechanism: symbolic compression stays close to the learned neural surrogate and inherits its constitutive bias rather than correcting it.

Appendix A.6. Restricted symbolic families

The symbolic stage is intentionally restricted. For diffusion, the candidate library contains polynomial families up to degree 3, a $(2, 2)$ rational family, an exponential-decay family, and a saturation family. For reaction, the candidate library contains polynomial families up to degree 4, a $(2, 2)$ rational family, a $u\exp(\cdot)$ family, and a $u/(1 + \cdot)$ saturation family. The selected expression is the lowest-complexity candidate whose fit mean-squared error lies within a 5% relative tolerance and a 10^{-8} absolute tolerance of the best candidate. This library is not meant to be exhaustive; it is a controlled interpretable compression family.

Appendix A.7. Reproducibility notes

The main result tables are generated from CSV summaries written by the benchmark scripts in the accompanying codebase, including symbolic-compression, excitation, and cross-resolution runs. Fixed random seeds are used throughout. Exact bitwise determinism across hardware is not claimed because PyTorch deterministic kernels are not globally forced, but repeated runs in the current environment reproduce the aggregate conclusions and the saved benchmark CSVs.

Appendix A.8. Code availability

The implementation, manuscript source, selected paper-facing table summaries, and figure assets are available at https://github.com/ToughClimb/closure-bias-inheritance.